

Name KEY

Total 12 points

Please show all your work.

1. (3 pts) A study of 350 patients with anxiety disorders was conducted to find a link between coffee drinking and anxiety disorders. Identify the population and the sample.

Population: Patients with anxiety disorders.
Sample: The 350 patients that were studied.

2. (3 pts) Identify the data set level of measurement (nominal, ordinal, interval, ratio):

a) the years of birth for the runners in the marathon Ratio

b) a list of post office box numbers Nominal

c) the sizes for a sample of automobiles:

subcompact compact midsize large Ordinal

3. (3 pts) Identify the sampling technique used (random, cluster, stratified, systematic, convenience):

a) After an earthquake, a disaster area is divided into 180 equal grids. 25 of the grids are selected, and every occupied household in the grid is interviewed to help focus efforts on what residents require the most.

cluster

b) Chosen at random, 500 rural and 500 urban persons age 65 or older were asked about their health and experience with prescription drugs.

Stratified

c) A list of managers is compiled and ordered. After a starting number is randomly selected, every eleventh name is selected until 275 managers are selected. The managers are asked about the use of digital media.

Systematic

4. (3 pts) Decide which method of data collection you would use to gather data for the following study and explain your reasoning:

A study on the effect of low dietary intake of Vitamin C and iron on lead levels in adults

Experiment: You want to measure the effect
of the intake of Vitamin C
This changes the existing conditions

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1. (3 pts) When 1250 American households were surveyed, it was found that 57% of them owned two cars. Identify the population and the sample.

Population: American households
Sample: The 1250 American households that were surveyed.

2. (3 pts) Identify the data set level of measurement (nominal, ordinal, interval, ratio):

a) ages of people in chemistry class Ratio

b) list of ZIP codes in Utah Nominal

c) the rating of a movie ranging from "poor" to "good" to "excellent" Ordinal

3. (3 pts) Identify the sampling technique used (random, cluster, stratified, systematic, convenience):

a) To ensure customer satisfaction, every 21st phone call received by customer service will be monitored.

Systematic

b) To get the opinion from students, the reporter selected to interview 12 freshmen, 9 sophomores, 12 juniors, and 14 seniors from 150 freshmen, 113 sophomores, 150 juniors and 175 seniors at a certain high-school.

Stratified

c) At a big local community center there are 10 ESL (English Second Language) classes offered. A researcher interviews all of the students from 2 randomly selected classes.

Cluster

4. (3 pts) Decide which method of data collection you would use to gather data for the following study and explain your reasoning:

A study where 68 individuals were given a new medication and 68 individuals were given a placebo, and the effect of this new medication on lowering the blood pressure was measured.

Experiment: You want to measure the effect of the new medication vs the placebo. This changes the existing conditions